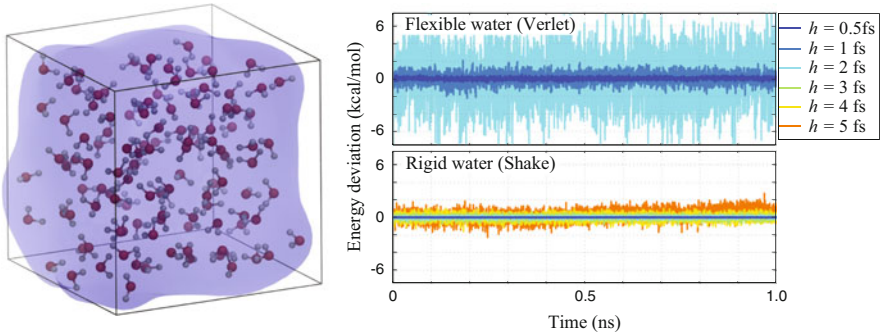


Fig. 4.5 We simulate a cubic box of TIP3P water (*left*) with box side length 16A using periodic boundary conditions. The deviation in the total energy is shown for simulations using different stepsizes (*right*), when using either the flexible or rigid (constrained) model. Note that the flexible model could not be stably simulated at the larger stepsizes

the timestep makes constrained simulation typical more efficient than unconstrained dynamics. In general, a doubling of the timestep is often possible.

A concern might be that the change of the model from unconstrained to constrained could alter the properties of the system in a way that would affect the overal simulation results. In practice, it is normally found that the other model and sampling errors present mean that the errors due to using constraints are relatively minor, particularly if the constraints are limited to bond lengths only, and even less of a factor if the constraints are limited to the bonds between hydrogen atoms and heavier atoms in biomolecules.

4.4 Rigid Body Systems

In some cases the effect of introducing constraints may be to rigidify completely certain molecular groups. Constraints are also used as part of coarse-graining procedures in which large collections of atoms may have their relative motions eliminated. In this case it is often useful to replace the rigid groups by ideal rigid bodies and to model these using rigid body dynamics. There is an extensive research literature on numerical methods for rigid body molecular dynamics, our discussion here of this topic is limited to a few schemes that we feel offer a reasonable combination of efficiency and geometric fidelity. For a more complete introduction to the topic, we recommend [80, 84] and the references therein.

As an example, in the TIP4P model for water, one of the most popular, the waters are assumed to be rigid bodies and there are 3 charge sites associated to each: the locations of the hydrogen nuclei and an additional “ghost” site located in the plane created by the three nuclei and along the perpendicular bisector of the HOH angle (Fig. 4.6). The potential energy for the interaction of a pair of such molecules, the

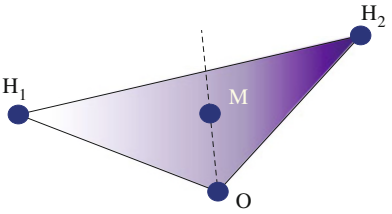


Fig. 4.6 Rigid representation of a water molecule used in TIP4P, indicating the "ghost" charge site

sets of whose charge sites are denoted by A and B, takes the form

$$\varphi^{\text{water}-\text{water}} = \sum_{i\in A}\sum_{j\in B} k_1 \frac{Q_i Q_j}{r_{ij}} + \frac{k_2}{r_{\text{OO}}^{12}} - \frac{k_3}{r_{\text{OO}}^{6}},$$

where r_{OO} represents the distance between the oxygen atoms of the two molecules, and k_1, k_2, k_3 are positive constants. Similar potentials are needed for interactions between the atoms of the solute and the atoms of the water molecules.

More generally, consider a collection of N atoms (masses m_i, positions $\boldsymbol{q}_i$, $i = 1, 2, \ldots, \text{N}$) subject to an external potential U. The kinetic energy is

$$K = \frac{1}{2}\sum_{i=1}^{\text{N}} m_i \|\dot{\boldsymbol{q}}_i\|^2.$$

The atomic positions may be referred to a coordinate system placed at the center of mass of the collection

$$\boldsymbol{q}_{\text{cm}} = \frac{\sum_{i=1}^{\text{N}} m_i \boldsymbol{q}_i}{M},$$

where $M = \sum_{i=1}^{\text{N}} m_i$ is the total mass. We write $\boldsymbol{q}_i = \boldsymbol{q}_{\text{cm}} + \boldsymbol{\delta}_i$, where $\sum_{i=1}^{\text{N}} m_i \boldsymbol{\delta}_i = 0$. Thus the kinetic energy becomes

$$\begin{aligned} K &= \frac{1}{2}\sum_{i=1}^{\text{N}} m_i \|\dot{\boldsymbol{q}}_{\text{cm}}\|^2 + \sum_{i=1}^{\text{N}} m_i \dot{\boldsymbol{q}}_{\text{cm}} \cdot \dot{\boldsymbol{\delta}}_i + \frac{1}{2}\sum_{i=1}^{\text{N}} m_i \|\dot{\boldsymbol{\delta}}_i\|^2 \\ &= \underbrace{\frac{M}{2}\|\dot{\boldsymbol{q}}_{\text{cm}}\|^2}_{\equiv K_{\text{trans}}} + \underbrace{\frac{1}{2}\sum_{i=1}^{\text{N}} m_i \|\dot{\boldsymbol{\delta}}_i\|^2}_{\equiv K_{\text{rot}}}, \end{aligned}$$

where we have used the fact that $\sum_{i=1}^{\text{N}} m_i \boldsymbol{\delta}_i \equiv 0$, hence that the time derivative also vanishes, in order to remove the middle term. At time t, each vector $\boldsymbol{\delta}_i$ can be related to its initial value by a certain orthogonal 3×3 rotation matrix $\boldsymbol{\Theta}(t)$: